Q1. Write a C program to accept the marks obtained by a student in 5 courses and calculate the average and percentage. (Consider the maximum marks in a course is 100).

CODE:

C

```c
#include <stdio.h>

int main()
{
    int m1, m2, m3, m4, m5;
    int total;
    float average, percentage;

    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

    total = m1 + m2 + m3 + m4 + m5;

    average = total / 5.0;

    percentage = (total / 500.0) * 100;

    printf("Total = %d\n", total);
    printf("Average = %.2f\n", average);
    printf("Percentage = %.2f\n", percentage);

    return 0;
}
```

OUTPUT:

Run

```
80 75 90 85 70
```

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.716 Mb

Sample Input

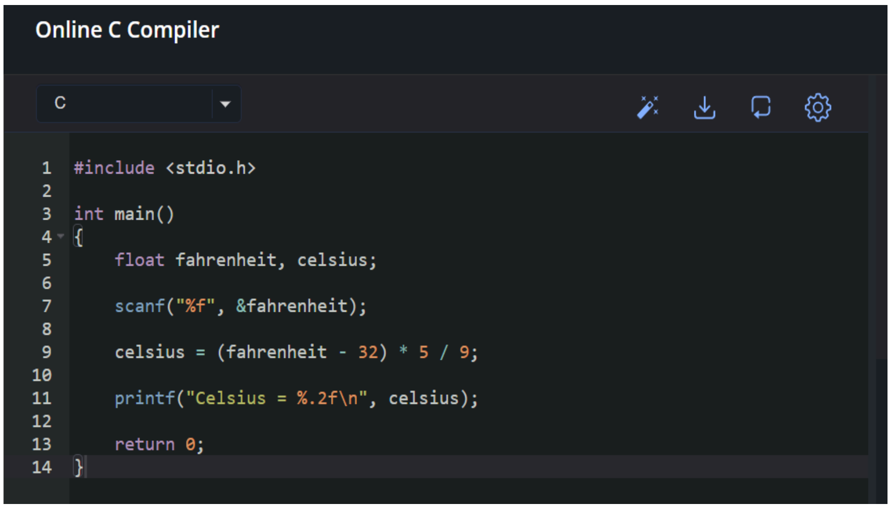
```
80 75 90 85 70
```

Your Output

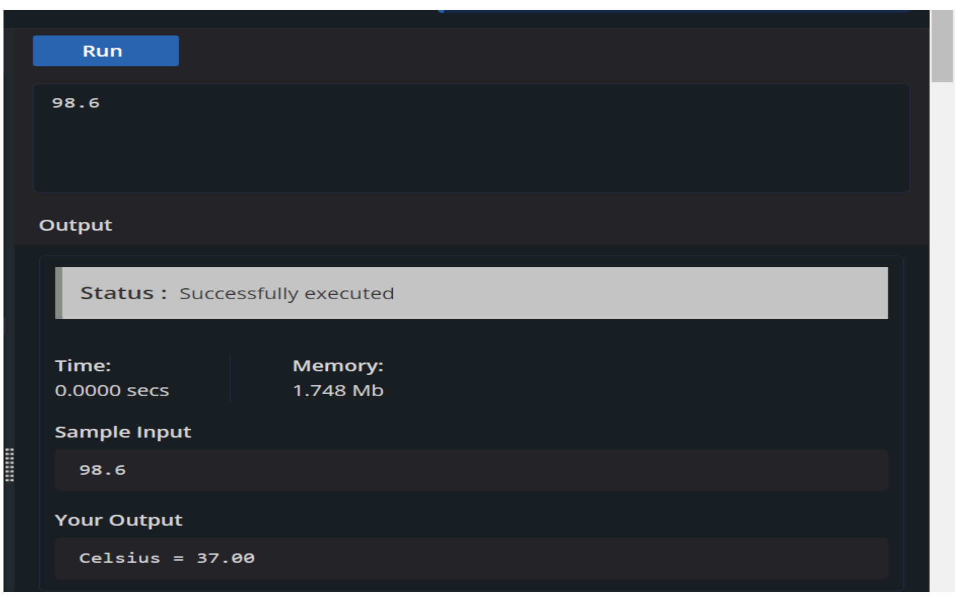
```
Total = 400
Average = 80.00
Percentage = 80.00
```

Q2. Write a C program that reads a temperature value in Fahrenheit from the user, converts it to Celsius using the formula celsius = (Fahrenheit - 32) * 5 / 9, and displays the converted temperature in Celsius

Code:



Output:



Q3. The total salary of any employee is a combination of basic, HRA, and allowance. The HRA is 40% of Basic. The basic allowance will be accepted by the user. Write a C program to calculate the total salary and display it.

Code:

C

```c
#include <stdio.h>

int main()
{
    float basic, allowance, hra, total;

    scanf("%f %f", &basic, &allowance);

    hra = 0.40 * basic;

    total = basic + hra + allowance;

    printf("Total Salary = %.2f\n", total);

    return 0;
}
```

Output:

Run

```
20000 5000
```

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.768 Mb

Sample Input

```
20000 5000
```

Your Output

```
Total Salary = 33000.00
```

Q4. Write a C program that reads the principal amount, rate of interest, and time in years from the user, calculates the simple interest using the formula simple_interest = (principal * rate * time) / 100, and displays the result.

Code:

C

```c
#include <stdio.h>

int main()
{
    float principal, rate, time, simple_interest;

    scanf("%f %f %f", &principal, &rate, &time);

    simple_interest = (principal * rate * time) / 100;

    printf("Simple Interest = %.2f\n", simple_interest);

    return 0;
}
```

Output:

Run

```
10000 5 2
```

Output

Status : Successfully executed

Time: 0.0000 secs

Memory: 1.704 Mb

Sample Input

```
10000 5 2
```

Your Output

```
Simple Interest = 1000.00
```